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TATA POWER-DDL
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CENTRAL ELECTRICITY AUTHORITY

DEPARTMENT OF
SCIENCE & TECHNOLOGY

AIDA
Collaboration | Innovation | Excellence

India
SMART UTILITY
Week 2026
10 – 14 March 2026, New Delhi

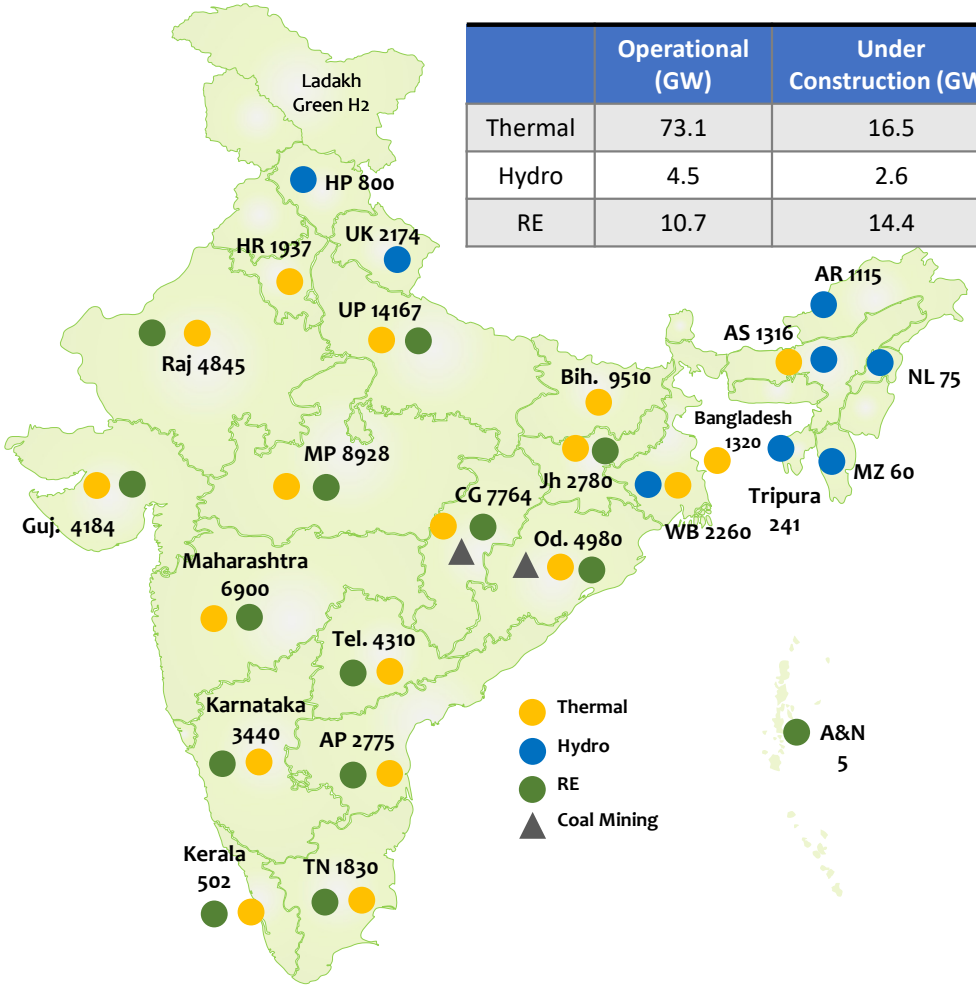
Session:

ROLE OF SMALL MODULAR REACTORS (SMR) IN THE NET-ZERO POWER SYSTEMS

Presented By

Arnada Prasad Samal, CEO, NPUNL & CGM NTPC Ltd

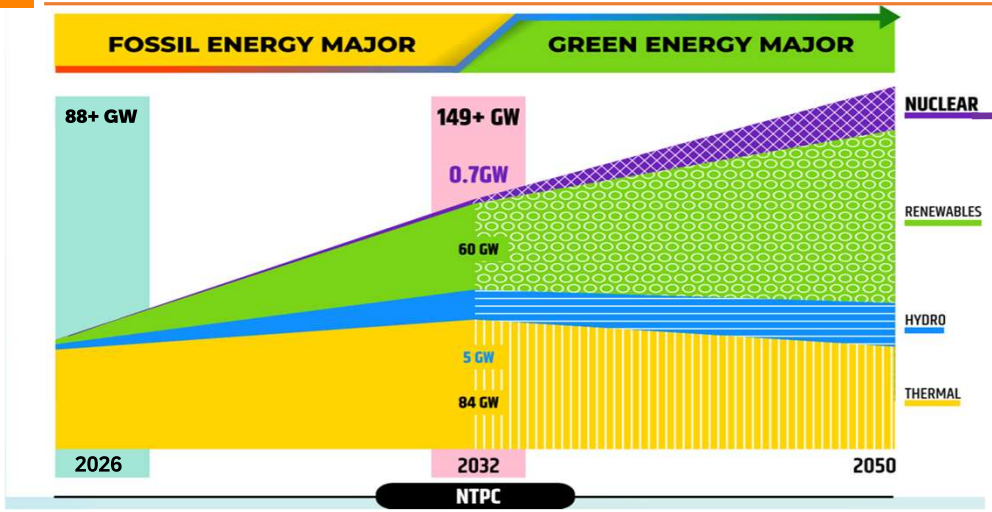
NTPC Group Profile



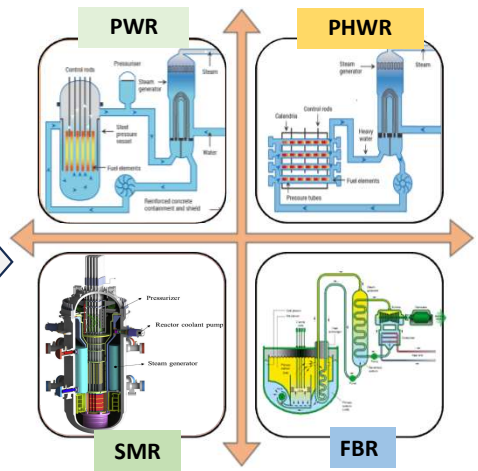
	Operational (GW)	Under Construction (GW)
Thermal	73.1	16.5
Hydro	4.5	2.6
RE	10.7	14.4

121.8 GW	Portfolio of Thermal , Hydro & Renewable Assets
88.27 GW	Capacity under Operation
33.56 GW	Capacity under Construction
10.73 GW	Renewable Capacity under Operation
~45 MMTPA	Coal Mining Capacity [6 Mines Operational + 3 Under Development]
27 Companies	11 Subsidiaries 16 Joint Ventures
Pan India Customers	Supplying Power to beneficiaries in all States & UTs
17,000+	Employees

NTPC Pursuing Energy Transition



Aiming
30 GW
Nuclear Capacity By 2047
Inline with GoI's
100 GW
Nuclear Roadmap



ANUSHAKTI VIDHYUT NIGAM LIMITED (ASHVINI)

A joint venture

- Re-Activation of ASHVINI — Dec, 2022
- GoI Approval to Build, Own & Operate NPP — Sept, 2024
- Approval of First Project MBRAPP — Sept, 2024
- Siting Consent — May, 2025
- Foundation laying of MBRAPP — Sept, 2025

MAHI BANSWARA RAJASTHAN ATOMIC POWER PROJECT (4 X 700 MW)

Excavation application submitted

- Siting Consent
- Foundation Stone laying
- Excavation Consent
- First Pour of Concrete
- First Approach To Criticality

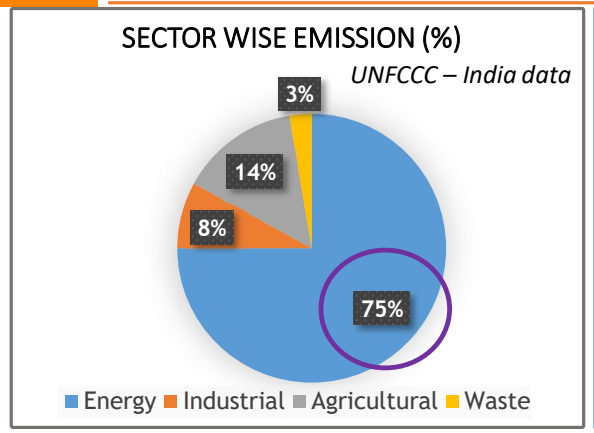
NTPC PARMANU URJA NIGAM LIMITED (NPUNL)

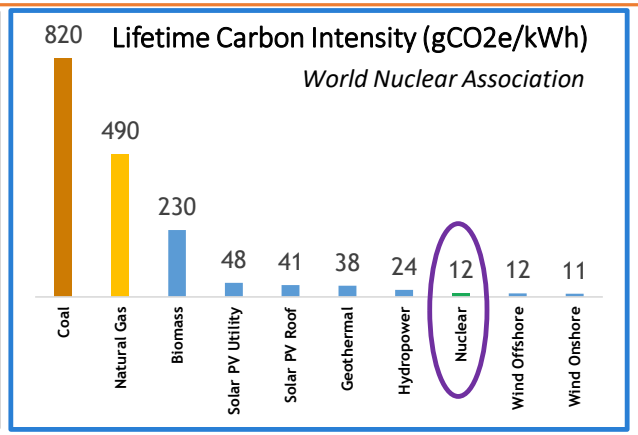
A Wholly Owned Subsidiary

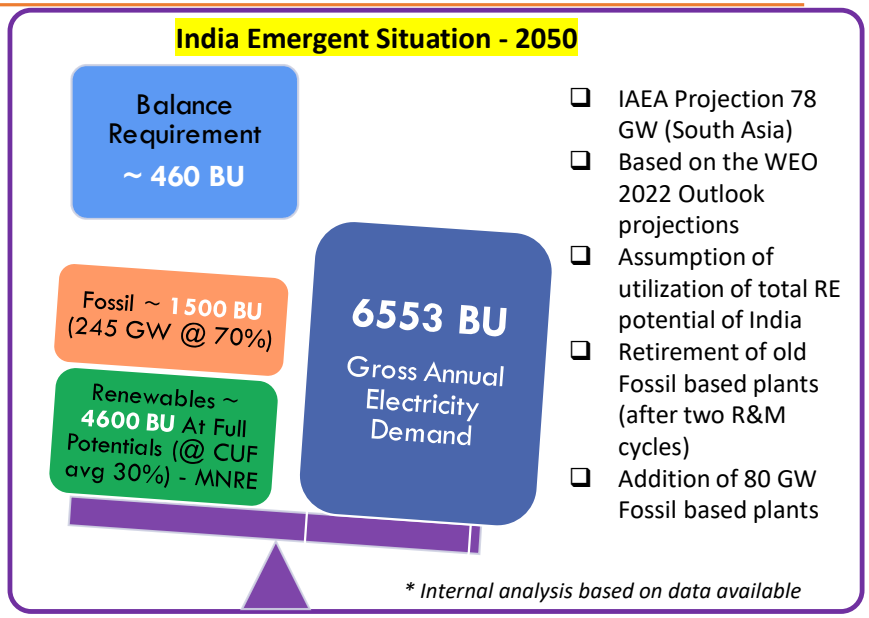
- Management Approval — May, 2024
- DIPAM Approval — July, 2024
- NITI AAYOG Clearance — Sept, 2024
- Approval by Ministry of Power — Sept, 2024
- Incorporation of NPUNL — Jan, 2025

Application submitted to DAE to authorize Build, Own & Operate NPP

Need for Decarbonization & Nuclear Energy







Energy Storage systems have limitations.

Hence, requirement for dispatchable power to supplement renewable sources.

Globally, IEA & IAEA have expressed this.

INDIA's PANCHAMRIT

- To increase non-fossil fuel energy capacity to 500 GW by 2030.
- Increase the share of renewables in the energy mix to 50% by 2030.
- Reduce emissions by 1 billion Tonnes of CO2.
- Reduce the emissions intensity of its economy by 45%.
- Carbon Net-zero by 2070

Nuclear fits into 4 out of above 5 Amrits!

Potential RE by MNRE

Solar : 850 GW (Roof Top Solar 102 GW)

Wind : 765 GW (Offshore 70 GW)

Small Hydro : 21 GW *Biomass : 20 GW* ***Waste*** *: 5.7 GW*

India's Present Electricity consumption 1824 BU (2024-25)

Expected Electricity Consumption 6553 BU (2050) *estimated by WEO 2022*

Total renewable energy potential 4600 BU (30% CUF)

Nuclear energy is the only option to provide the low carbon dispatchable power – recognized worldwide

National inclination towards Nuclear Energy

BUDGET 2025-26

Promotion of Nuclear

- **Nuclear Capacity target 100 GW by 2047**
- **Nuclear Energy Mission for SMR**
- **5 Indigenous SMRs operational by 2033**
- **Outlay of 20,000 crore**
- **Policies** *SHANTI Act enabling wider participation for the planned capacity addition*

✓ Committee constituted by GoI for "**100 GW Nuclear Roadmap**"

✓ Committee submitted report with the following capacity plan*

	PHWR	LWR	FBR	SMR / BSR / BSMR	Total
Operational & Under implementation	15.7	6.3	0.5	-	22.5
DAE Units	14	17.6	4	-	32.6
Other PSUs	16.8	15	3.5	10	45.3
Gross	46	39	5	10	~ 100

*Report Reference: https://cea.nic.in/wp-content/uploads/notification/2025/10/Roadmap_Final_30062025.pdf

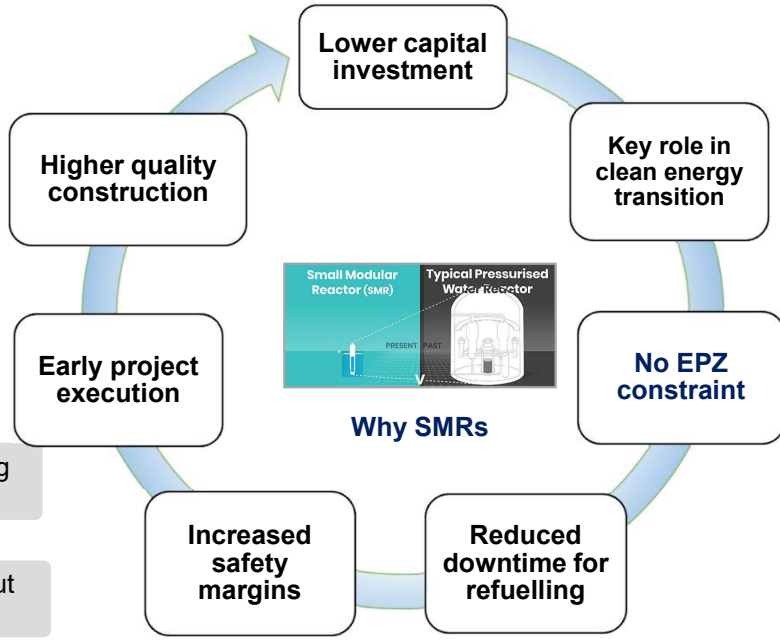
Net Zero Scenario		
Indicator	2050	2070
Electricity Consumption (TWh)	8100	13000
Nuclear Capacity (GW)	95-105	**290-320**

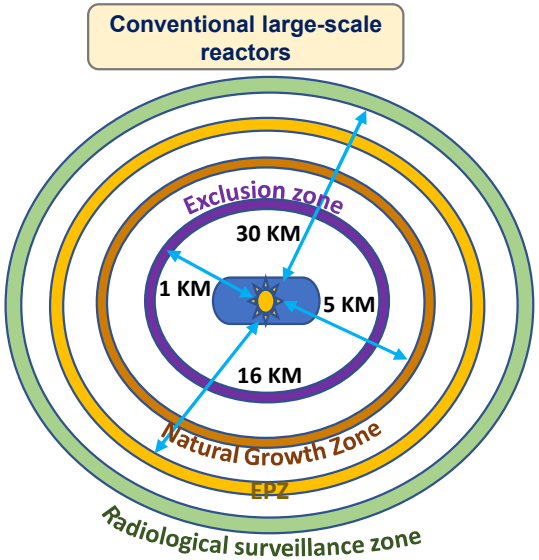
3x

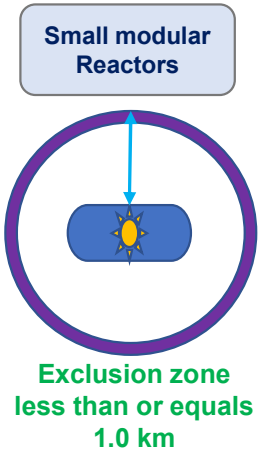
Reference: A STUDY REPORT ON SCENARIOS TOWARDS VIKSIT BHARAT AND NET ZERO (2026) by NITI Aaayog:https://niti.gov.in/publications/division-reports

Small Modular Reactor

- Nuclear Reactors with a power output between 10 MWe and 300 MWe
- Module factory fabrication, factory-based construction in order to maximise economies of series
- The different modules can then be transported and assembled onsite
- Predictability and savings in construction times
- Level of construction quality and efficiency improvement







Passive cooling system & infinite cooling through design

↓

Ensures core cooling for > 7 days without operator intervention

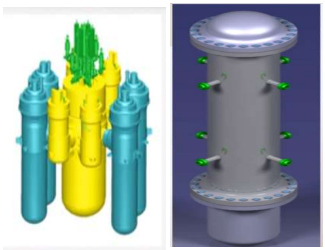
↓

Rules out core meltdown & reduces core damage frequency drastically

Types:

Block: Each of the major components of the reactor, i.e. SG, pressuriser & pumps are welded directly to RPV

Integral: The reactor and associated components like SG, Pumps & Pressurizer are all positioned within a single pressure vessel



Global Status of SMR

Operational	Under construction	Basic/detailed design stage	Conceptual design stage
4 KLT-40S (2), HTR PM (2)	3 CAREM (2028), ACP100 (2026), BRESTOD (2026), Hermes (*)	67	28

*USA – Low Power demonstration reactor (Fluoride salt based) – Heat energy (not power production)

Fuel Cycle

Most SMR designs – HALEU (5% to 20% U-235)
HALEU commercial stage production in Russia & USA

Type	Fuel used	Refuelling cycle
Land based Water Cooled Reactors	LEU (< 5%)	18–24 months
Marine based Water Cooled Reactors	HALEU	Up to 120 months
Gas Cooled Reactors	LEU / HALEU (TRISO coated)	25-60 months
Molten Salt Reactors	LEU / HALEU (TRISO coated)	Up to 150 months
Liquid Metal M Fast Neutron Reactors	HALEU (14-20%)	Up to 30 years

SMR in Indian Context

1

BSMR-220

DPR approved, financial sanction awaited, pre-project activities are under progress

2

SMR-55

Design detailing under progress

3

5MWth HTGR

H2 generation, DPR prepared, financial & administrative approval awaited

Department of Atomic Energy

Azadi Ka Amrit Mahotsav

PARLIAMENT QUESTION: DEVELOPMENT OF SMALL MODULAR REACTOR

Posted On: 04 FEB 2026 6:26PM by PIB Delhi

Nuclear energy is envisaged a clean and reliable energy source. Small Modular Reactors (SMRs) are prioritized for deployment in brownfield site as captive power plants for rapid decarbonization of energy intensive sectors, repurposing of retiring fossil fuel based power plants and deployment in remote location with no grid connectivity for catering to energy requirements.

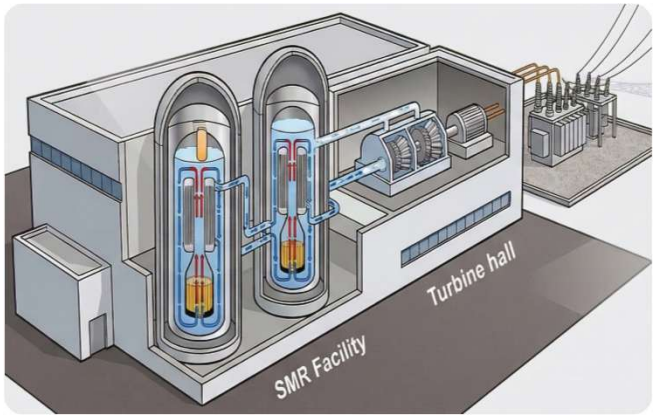
a. SMR is a promising technology in industrial de-carbonization especially where there is a requirement of reliable and continuous supply of power. SMRs can be deployed as captive plants for large industries repurposing of retired fossil fuel based power plants. As the exclusion zone required will be significantly lower due to enhanced safety, this, reduces the land requirement. They are suitable candidates for setting up in retired power plants, off grid areas also. SMRs can be engineered to have load follow characteristics for increasing flexibility of operation. SMRs, in general, has lower construction time and hence lower capital cost. Their design can be standardized for serial production.

b. Under the Nuclear Energy Mission, Bhabha Atomic Research Centre (BARC), a Constituent Unit of Department of Atomic Energy (DAE) has recently initiated design and development of SMRs namely :

1. 220 MWe Bharat Small Modular Reactor (BSMR-200). The detailed project report has been approved and financial sanction is awaited. Various pre-project activities are under progress.
2. 55 MWe Small Modular Reactor (SMR-55). The proposal has been approved in- principle and design detailing is in progress. Design and development of major prototype equipment are in progress.
3. Up to 5 MWth high temperature gas cooled reactor meant for hydrogen generation. The detailed progress report (DPR) has been prepared and financial and administrative approval is being sought and design detailing is in progress.

Application of SMR

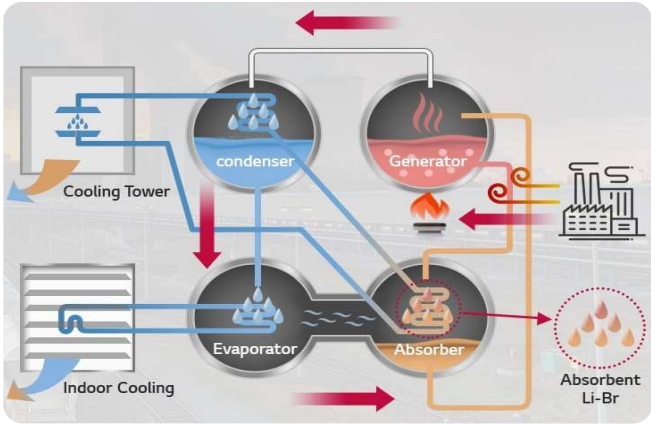
Power Generation



Industrial Heating

Textile Refinery **Chemicals Food Process** **Fertilizers Paper**

Cooling & Refrigeration



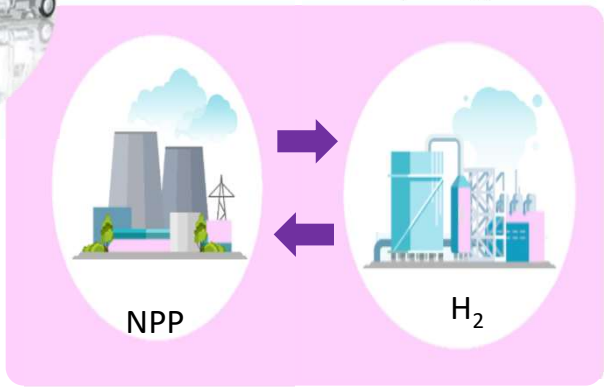
Desalination & Water Grid

Distributed generation

Micro-grid applications

SMR in remote locations

Nuclear Hydrogen



- Low Temp (LTE) (50-60 kWh/m3)
- High Temp (HTSE) (42-44 kWh/m3)
- Thermochemical ($S\text{-}I_2$, $Cu\text{-}Cl_2$ Cycles T > 600°C)

NTPC SMR development till date & way forward

Foreign Collaborations

HOLTEC INTERNATIONAL	MoU Signed with Holtec International Discussion underway for identifying suitable mode of collaboration
ROLLS ROYCE SMR	Initial dialogue with Rolls Royce for deployment

Constraint in Immediate Deployment

- **Regulatory Void:** Absence of a specific regulatory framework for SMRs in India, reflecting the early development stage of this technology.
- **Technology Maturity:** Expected by 2033
- **Suitability:** Suitable for brownfield development

Way Forward

1. Pursue a Dual Technology Strategy through indigenous and foreign collaboration route
2. Use SMRs for hard to Abate sector
3. Develop a Domestic SMR Manufacturing Ecosystem

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THANK YOU

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